Analytical MCQ Assessment

High-level cognitive evaluation paper.

SECTION A: Multiple Choice Questions

Q1. A network administrator wants to block incoming traffic based solely on source IP addresses and port numbers without tracking the state of active connections. Which type of firewall is most suitable for this task? (1 Marks)

A. Stateful Inspection Firewall
B. Packet Filtering Firewall
C. Application Gateway Firewall
D. Next-Generation Firewall

Q2. Which type of firewall operates at the Session layer of the OSI model and monitors the TCP handshaking process to determine if a requested session is legitimate? (1 Marks)

A. Packet Filtering Firewall
B. Circuit-Level Gateway
C. Application-Level Gateway
D. Stateful Inspection Firewall

Q3. An organization requires a firewall that can inspect the actual content of HTTP requests to block specific SQL injection patterns. Which firewall type is best suited for this deep content inspection? (1 Marks)

A. Packet Filtering Firewall
B. Circuit-Level Gateway
C. Application-Level Gateway (Proxy)
D. Stateful Inspection Firewall

Q4. How does a stateful inspection firewall determine whether to allow an incoming packet that does not match an explicit rule? (1 Marks)

A. It automatically blocks all incoming packets that do not have an explicit rule.
B. It checks if the packet is part of an active, established outbound connection recorded in its state table.
C. It performs deep packet inspection on the payload to guess the user's intent.
D. It forwards the packet to a proxy server for further analysis.

Q5. A company wants to integrate intrusion prevention systems (IPS), application awareness, and deep packet inspection into a single security appliance. What type of firewall should they deploy? (1 Marks)

A. Stateful Inspection Firewall
B. Next-Generation Firewall (NGFW)
C. Circuit-Level Gateway

D. Packet Filtering Firewall

Q6. Which specialized firewall is specifically designed to protect web applications by filtering and monitoring HTTP traffic between a web application and the Internet? (1 Marks)

A. Web Application Firewall (WAF)
B. Next-Generation Firewall (NGFW)
C. Circuit-Level Gateway
D. Network Address Translation (NAT) Firewall

Q7. A security policy requires protecting individual endpoints from lateral movement attacks within the internal network. Which deployment type is most appropriate? (1 Marks)

A. Network-based Firewall
B. Host-based Firewall
C. Cloud-based Firewall
D. Perimeter Firewall

Q8. Which of the following is a primary limitation of a traditional packet filtering firewall? (1 Marks)

A. High computational overhead and latency
B. Inability to block traffic based on IP addresses
C. Vulnerability to IP spoofing because it does not verify connection state
D. Inability to operate at the Network layer of the OSI model

Q9. An enterprise is migrating its infrastructure to a multi-cloud environment and needs a centralized, scalable firewall solution managed entirely as a cloud service. What is this model called? (1 Marks)

A. Virtual Firewall
B. Firewall as a Service (FWaaS)
C. Host-based Firewall
D. Application-Level Gateway

Q10. Why might a network engineer choose a stateless packet filtering firewall over a stateful inspection firewall for a high-throughput backbone router? (1 Marks)

A. Stateless firewalls offer superior security and payload inspection.
B. Stateless firewalls require less memory and CPU because they do not maintain a connection state table.
C. Stateless firewalls can automatically detect application-layer attacks.
D. Stateless firewalls eliminate the need for access control lists (ACLs).

INSTRUCTOR ONLY - ANSWER KEY & RUBRIC

MCQ Solutions & Rationale

Q1. Correct Answer: B
Rationale: Packet filtering firewalls inspect packets individually based on header information like IP addresses and ports without maintaining connection state.

Q2. Correct Answer: B
Rationale: Circuit-level gateways operate at the Session layer to verify TCP handshakes and session legitimacy without inspecting packet payloads.

Q3. Correct Answer: C
Rationale: Application-level gateways inspect the application-layer payload of packets to detect and block specific malicious content.

Q4. Correct Answer: B
Rationale: Stateful firewalls keep track of active connections in a state table and automatically allow returning traffic associated with those established sessions.

Q5. Correct Answer: B
Rationale: Next-Generation Firewalls combine traditional firewall capabilities with advanced features like IPS, application control, and deep packet inspection.

Q6. Correct Answer: A
Rationale: A WAF is specifically tailored to protect web applications by analyzing layer 7 HTTP/HTTPS traffic for threats like cross-site scripting and SQL injection.

Q7. Correct Answer: B
Rationale: Host-based firewalls run locally on individual endpoints to secure them against internal threats and lateral movement.

Q8. Correct Answer: C
Rationale: Traditional packet filtering firewalls do not track connection states, making them susceptible to IP spoofing attacks.

Q9. Correct Answer: B
Rationale: Firewall as a Service (FWaaS) provides cloud-based, scalable network security managed centrally without physical hardware.

Q10. Correct Answer: B
Rationale: Stateless firewalls process packets faster and require fewer resources because they do not maintain state tables for active connections.